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Task Questions

1. Question 1

Answer:

Amazon EFS is a fully managed service that provides file system storage for Linux-based workloads. Multiple EC2 instances can be accessed. It also allows for a common interface, like a shared workspace or data source for workloads and application that run on more than one Amazon EC2 instance. In addition, it can also scale automatically from GBs to PBs of data without needing to provision storage.

Amazon EC2 instance store and Amazon EBS provide good choices for storing both root and data volume for a single instance. This is a block store. EBS volumes are independent storage units that can be attached to EC2 instances and used as primary storage for operating systems, databases, and other applications. So, the main difference between them is that EBS volumes are accessible to only one EC2 instance at a time unlike EFS. This property makes it suitable for instances that require low-latency access to block storage.

2. Question 2

Answer:

Amazon EFS uses the Network File System (NFS) version 4.x protocol. It can be used with any AMIs that support this protocol.

3. Question 3

Answer:

It is called **Amazon FSx (File System)**, it provides fully managed Windows file servers, backed by a fully-native Windows file system.

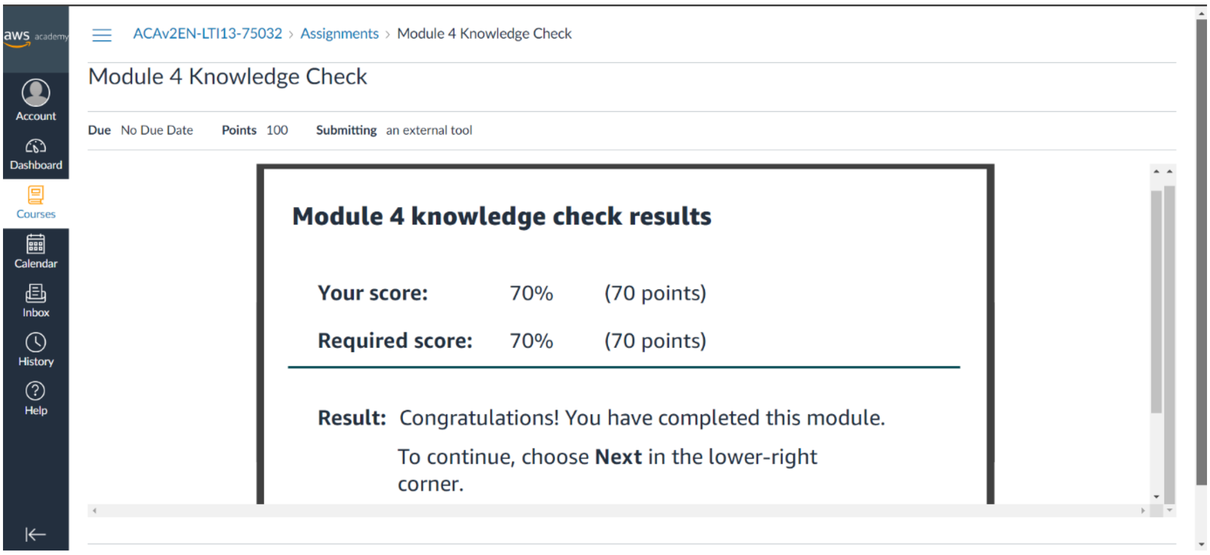
It uses the following protocols:

- New Technology File System (NTFS).
- Server Message Block (SMB).
- Distributed File System (DFS).
- Microsoft Active Directory, service that administers and organizes enterprise resources (including users, groups, and network resources).

For a data volume that serves multiple Microsoft Windows instances, use Amazon FSx for Windows File Server.

Quiz

1. Knowledge Check for Module 4



This knowledge check was comprised of the most important questions throughout the module 4. The questions asked was kind of a summary to the whole module 4. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

It was mostly about the EFS and EBS. What they are used for, their speed, their benefits and drawbacks, like the autoscaling and high availability are two of them. The security features they have. The file systems are managed for Linux by EFS whereas for Windows, Amazon FSx is used to manage the file systems. Then the pricing options for EC2 are asked as questions, they usually depend on the on-demand instances and spot instances. This is some of the

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(Amazon EFS)

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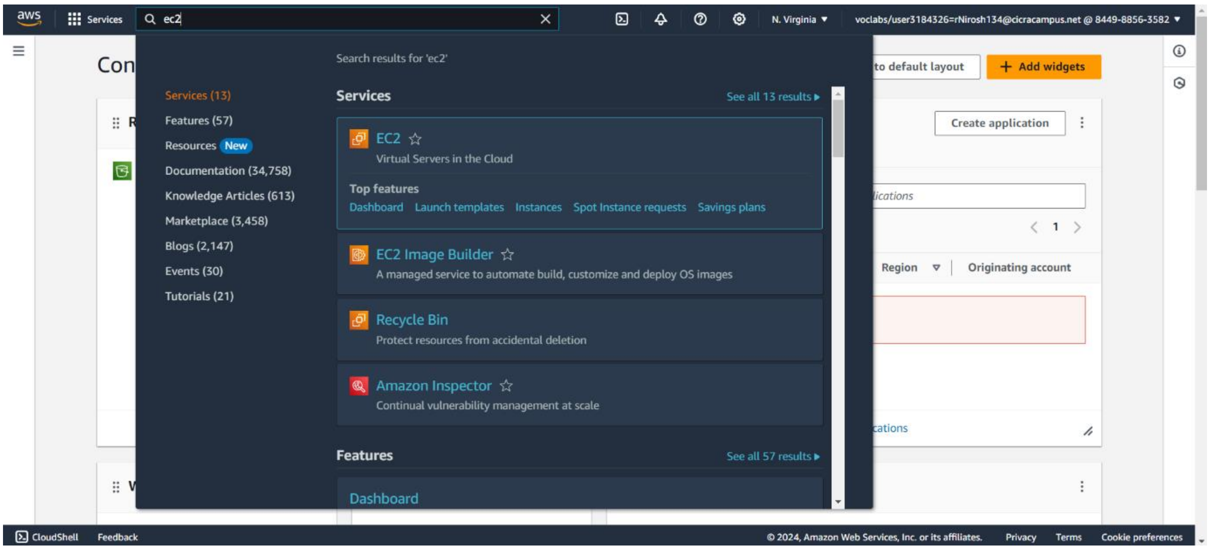
Module 4 Guided Lab - Introducing Amazon Elastic File System (Amazon EFS)

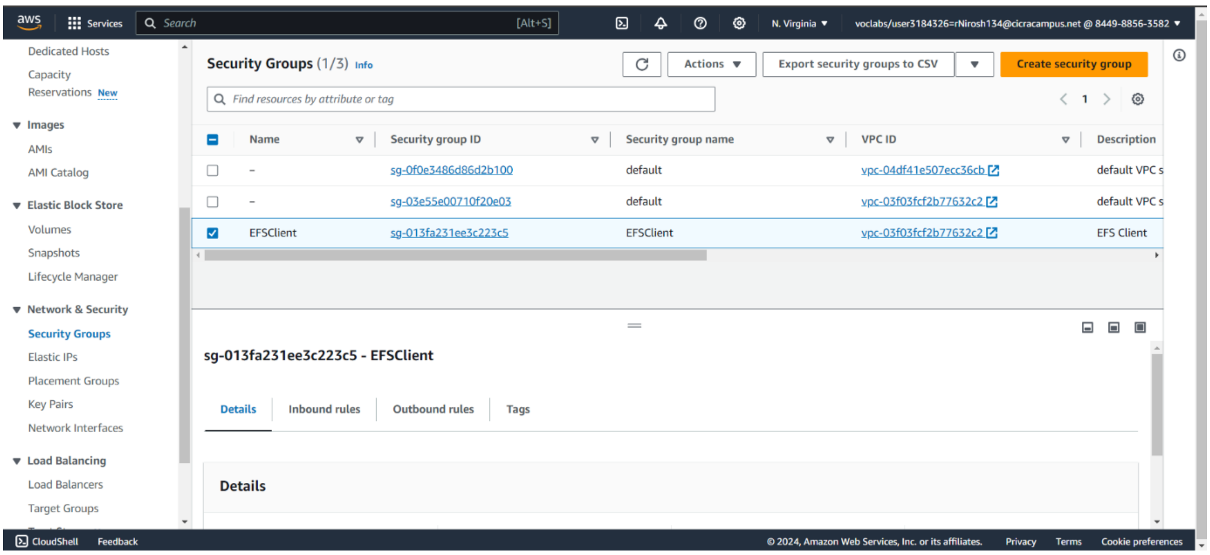
Task 1: Creating a security group to access your EFS file system

In this task I'll be setting up a security group in AWS to regulate inbound access on TCP port 2049 for NFS which is crucial for managing EFS mount targets.

First, I started by accessing the AWS Management Console and navigating to the EC2 service, where we can manage security groups.

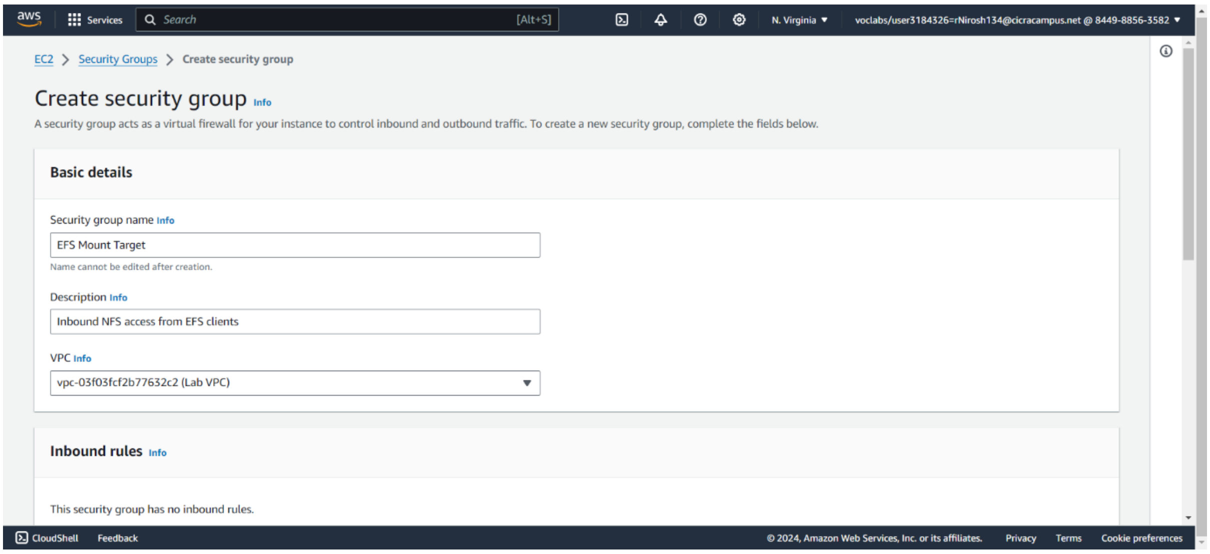


Within the EC2 service, we navigate to the Security Groups section. Security groups act as virtual firewalls that control the traffic for the instances.



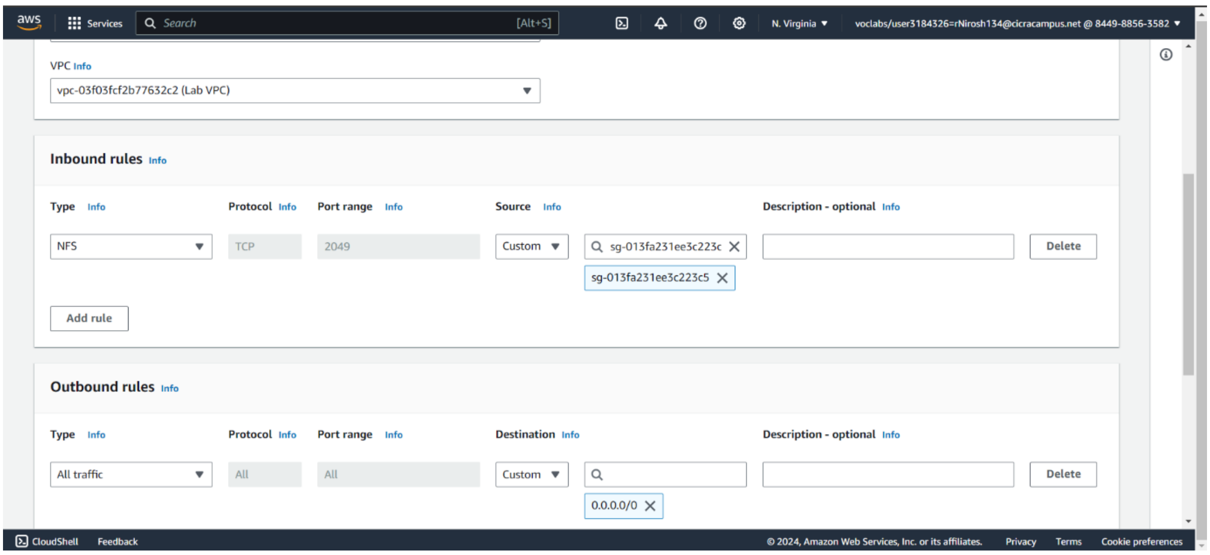
Identify an existing security group called "EFSClient" and copy its Security Group ID.

Created a new security group named "EFS Mount Target" specifically to manage inbound NFS access from EFS clients.

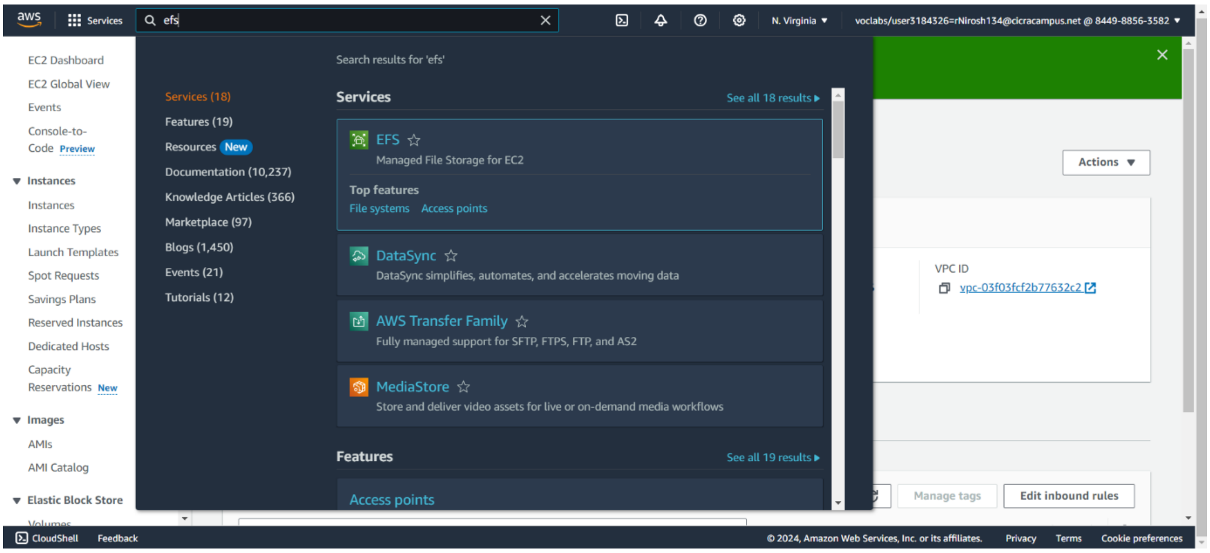


Then name and description given for the security group and specifying VPC.

An inbound rule is set up within the new security group to allow NFS traffic. The source will be specified as the "EFSClient" security group



Finally, a security group is set up with the specified the configuration.



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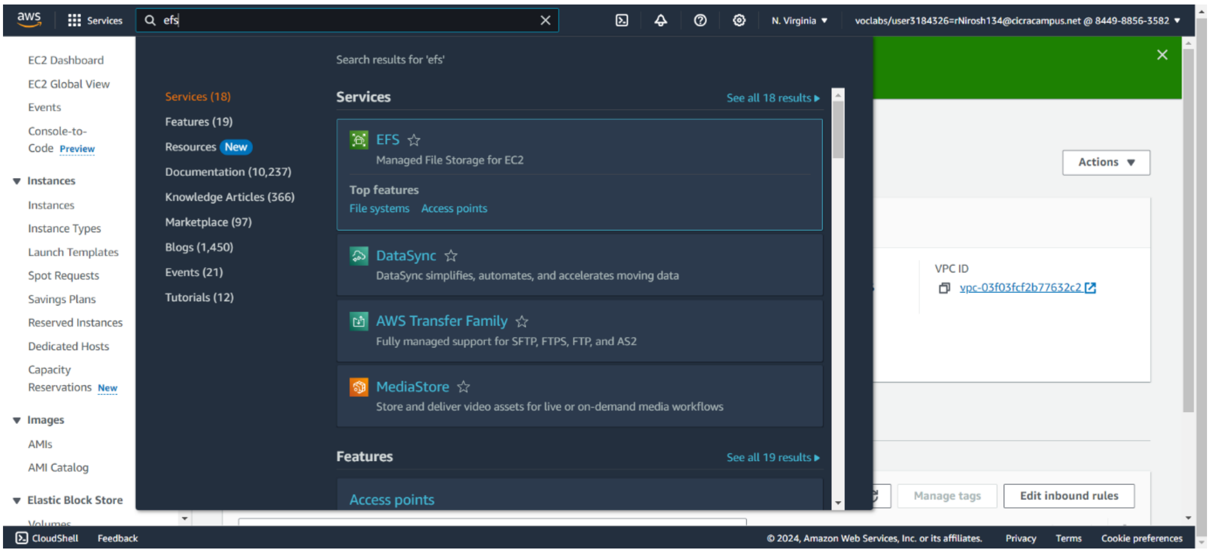
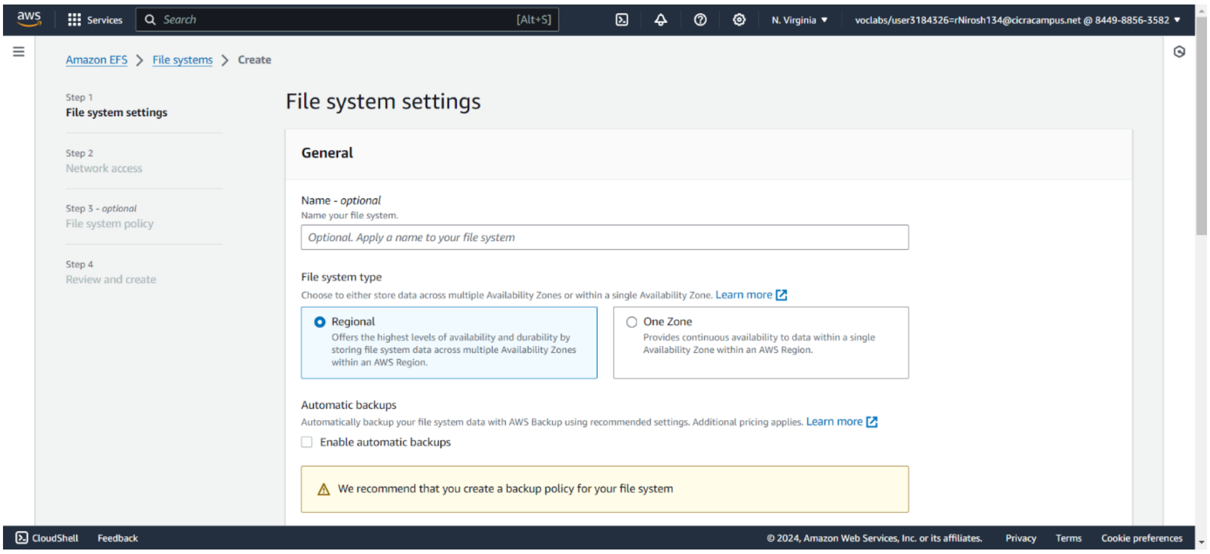
Task 2: Creating an EFS file system

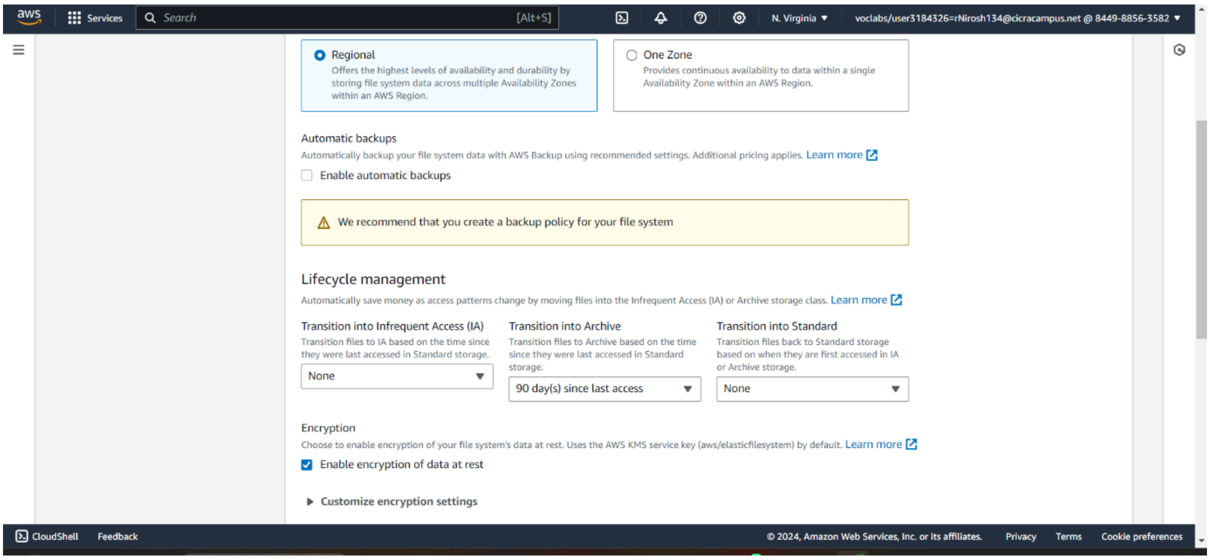
I'm setting up an Elastic File System (EFS) in AWS. I create a new file system, customize its settings, specify the VPC where it will be used, configure security groups for mount targets, review the configuration, and create the file system. Finally, I confirm its creation and wait for it to become available, ensuring secure access for EC2 instances within my chosen VPC.



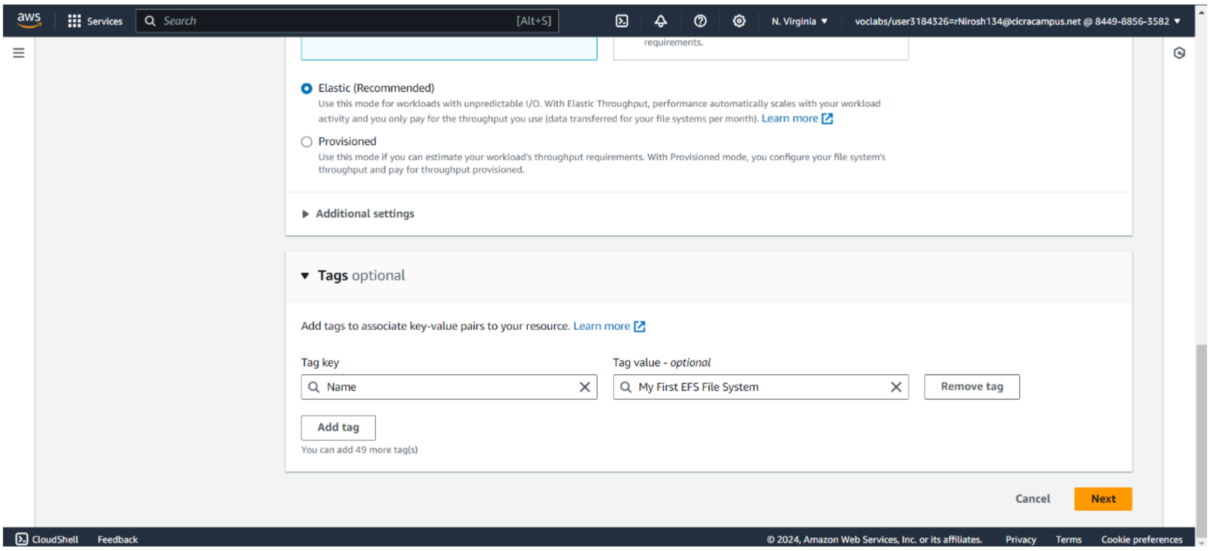
I navigate to the EFS service in the AWS Management Console and select "Create file system" to initiate setting up a new file system.



I choose "Customize" to specify detailed settings. In Step 1, I uncheck automatic backups and select "None" for lifecycle management.



I also tag the file system, naming it "My First EFS File System".



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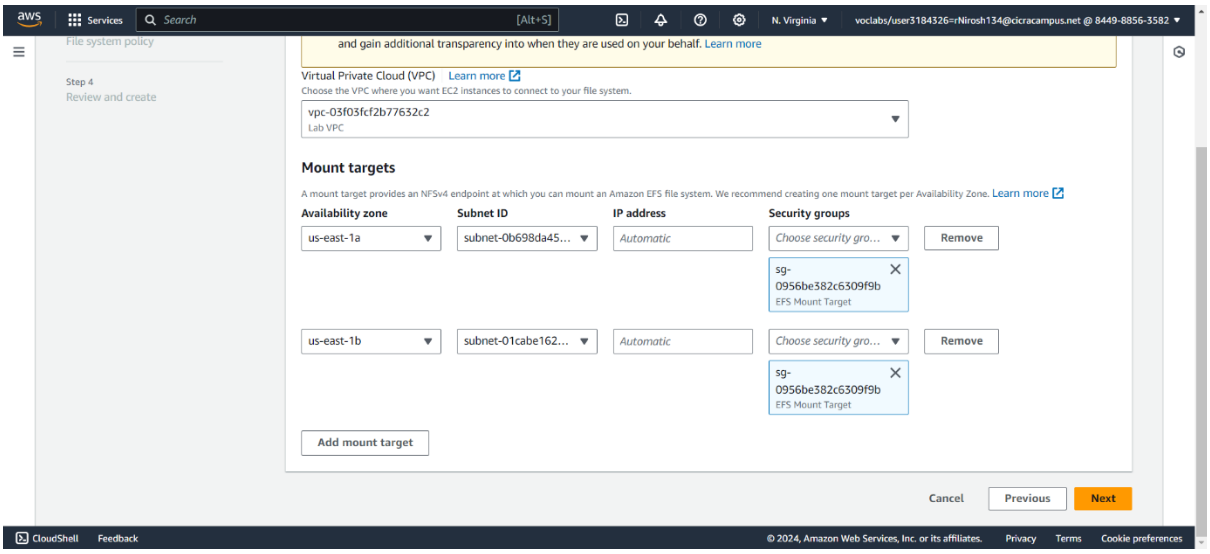
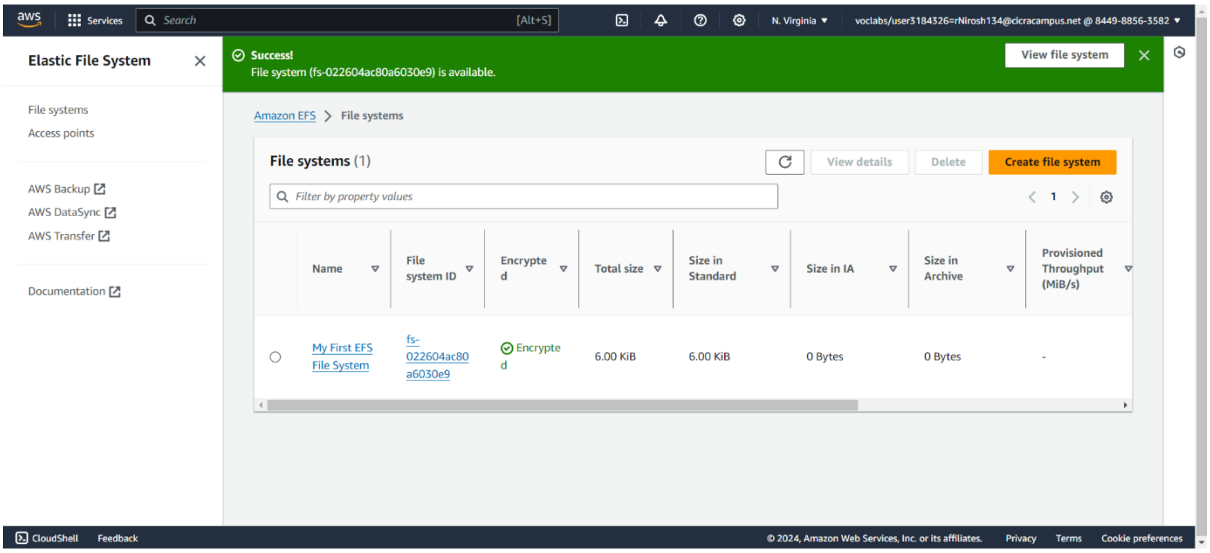
Selected the desired VPC "Lab VPC". Then removed the default security group from each Availability Zone's mount target and attach the "EFS Mount Target" security group to each one.



I confirm the configuration and choose "Create" to create the file system.

Shortly after creation, the file system state changes to "Available", followed by the mount targets' state a few minutes later. I wait for the mount target states to change to "Available" before proceeding further.



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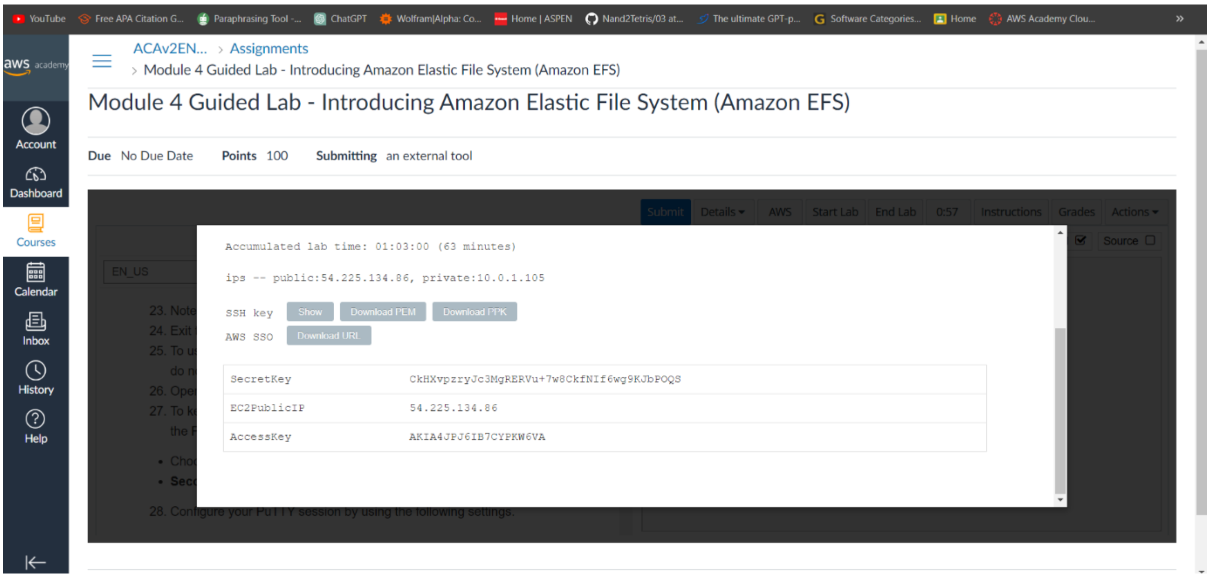
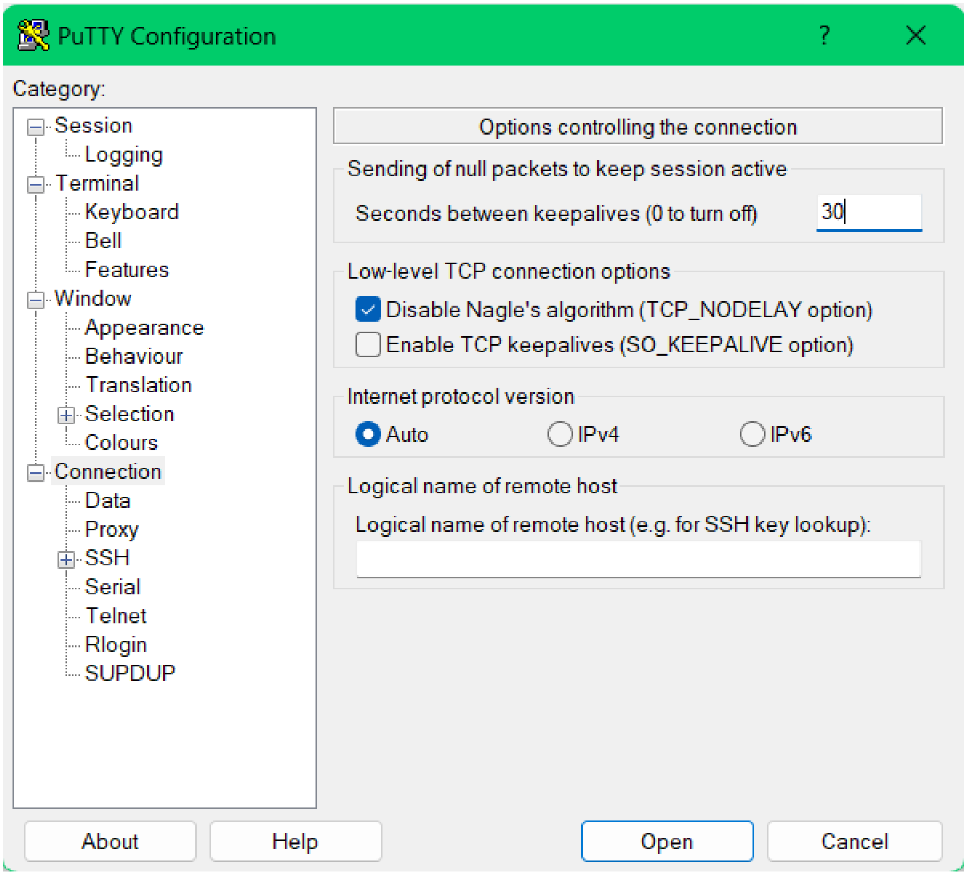
Task 3: Connecting to your EC2 instance via SSH

Opened the Details dropdown menu and select "Show" to reveal a Credentials window. Chose the "Download PPK" button and save the labsuser.ppk file to my computer. Then saved the EC2PublicIP address if it's displayed.



Opened the putty.exe, a terminal emulator application and configured PuTTY to keep the SSH session open for a longer period by setting the timeout to 30 seconds.



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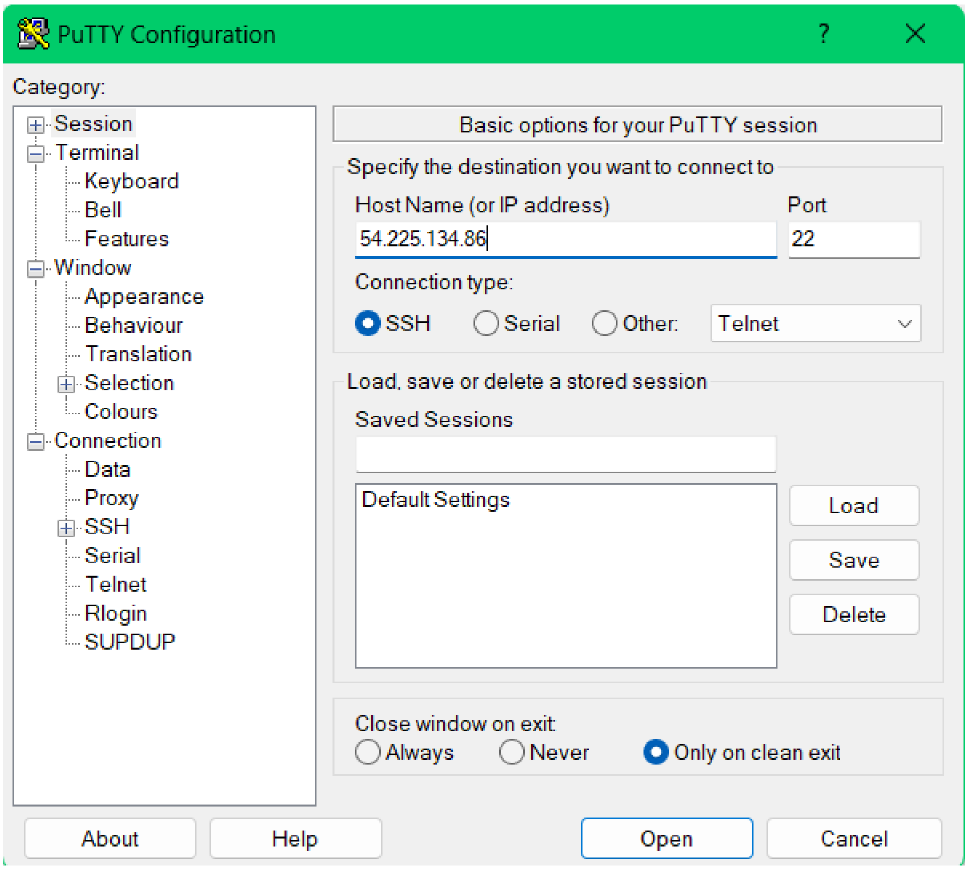
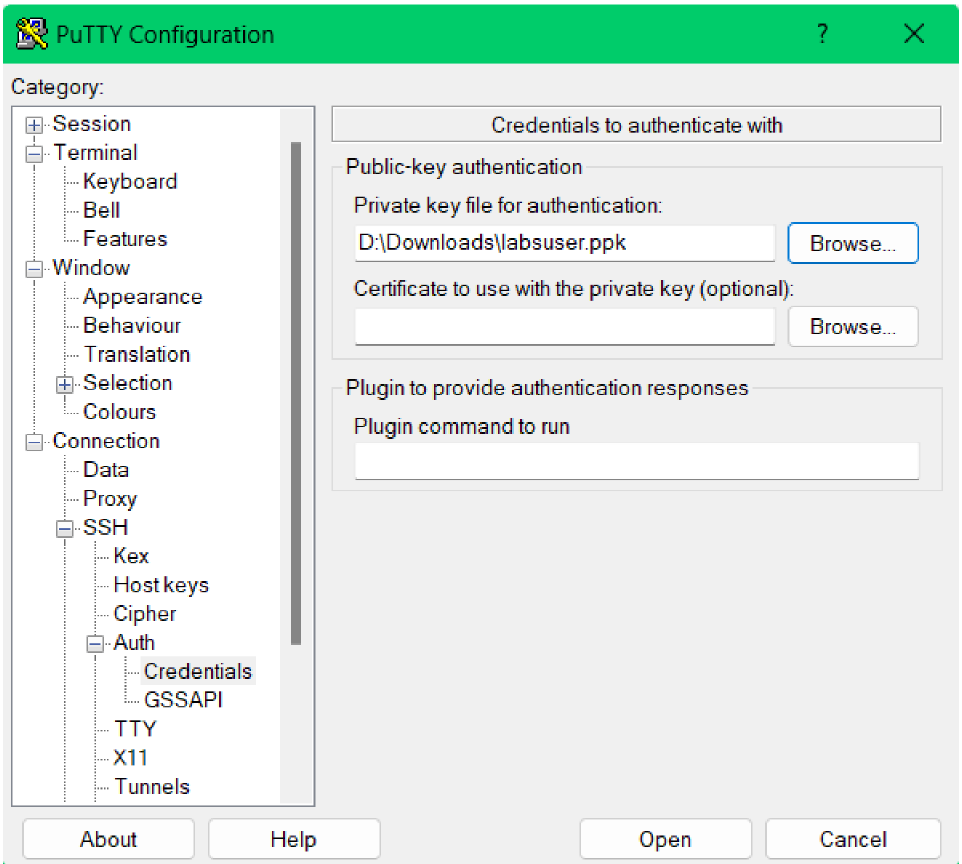
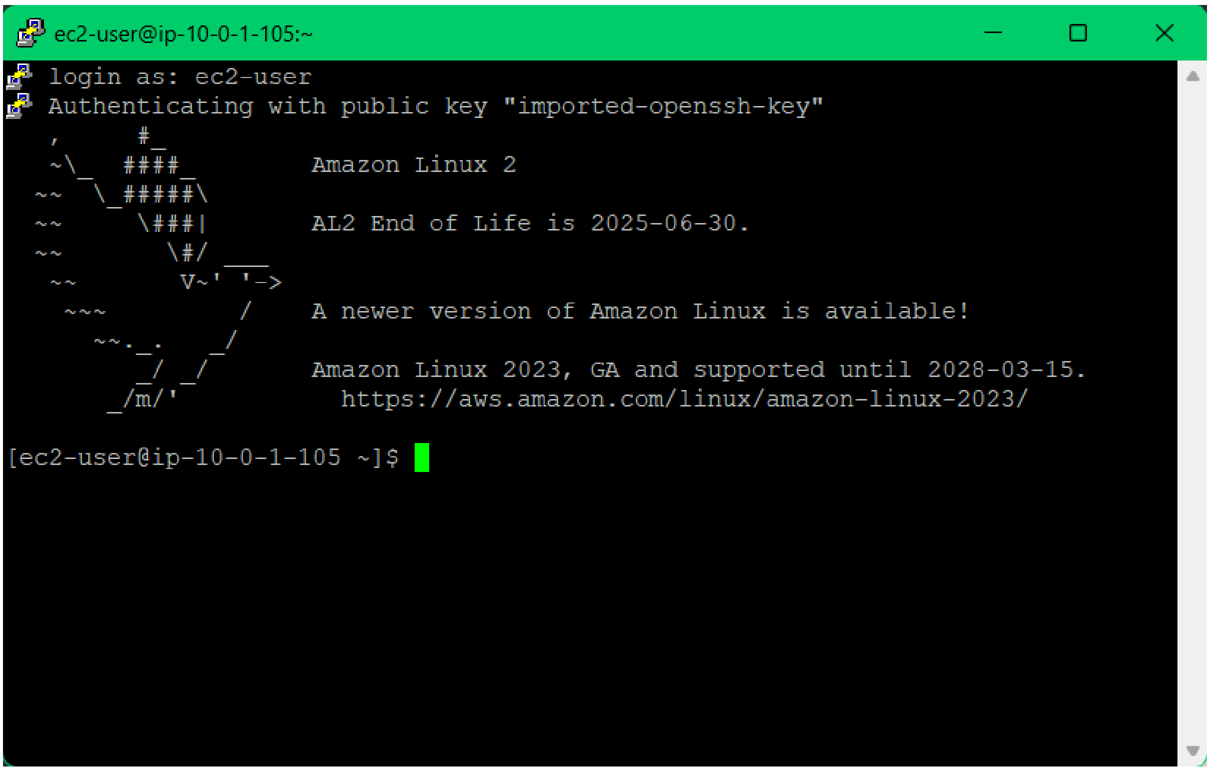
I copied the IPv4 Public IP from the Amazon EC2 console and paste it into PuTTY.

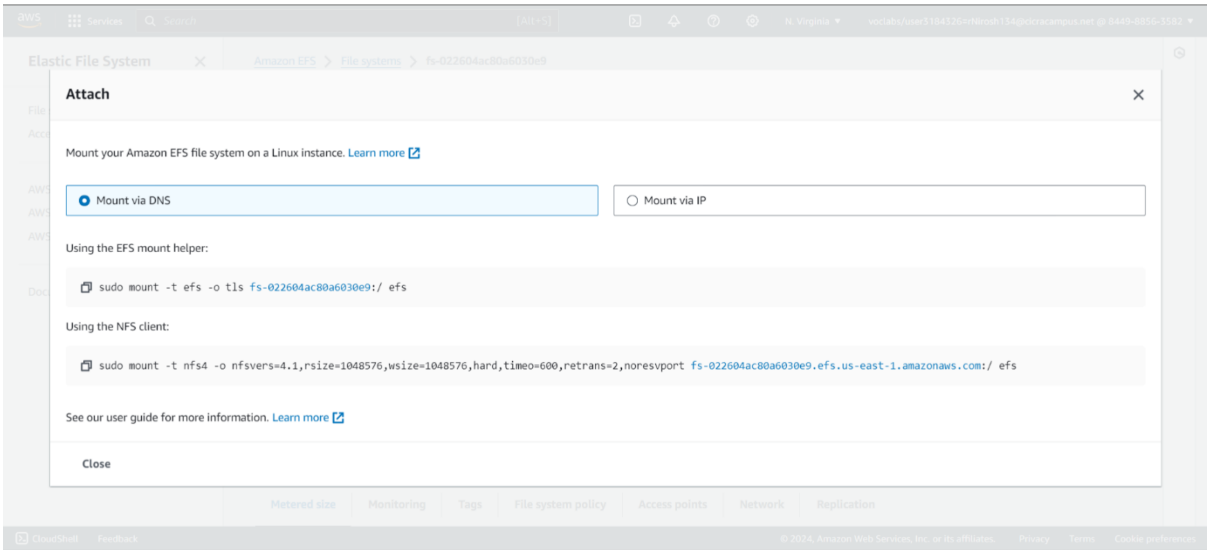


I navigate to PuTTY's SSH authentication settings and specify the labsuser.ppk file that I downloaded earlier.



I open the SSH connection to the EC2 instance. PuTTY prompts me to accept the host key to establish trust. Once connected, I'm prompted to log in as "ec2-user" to access the EC2 instance.

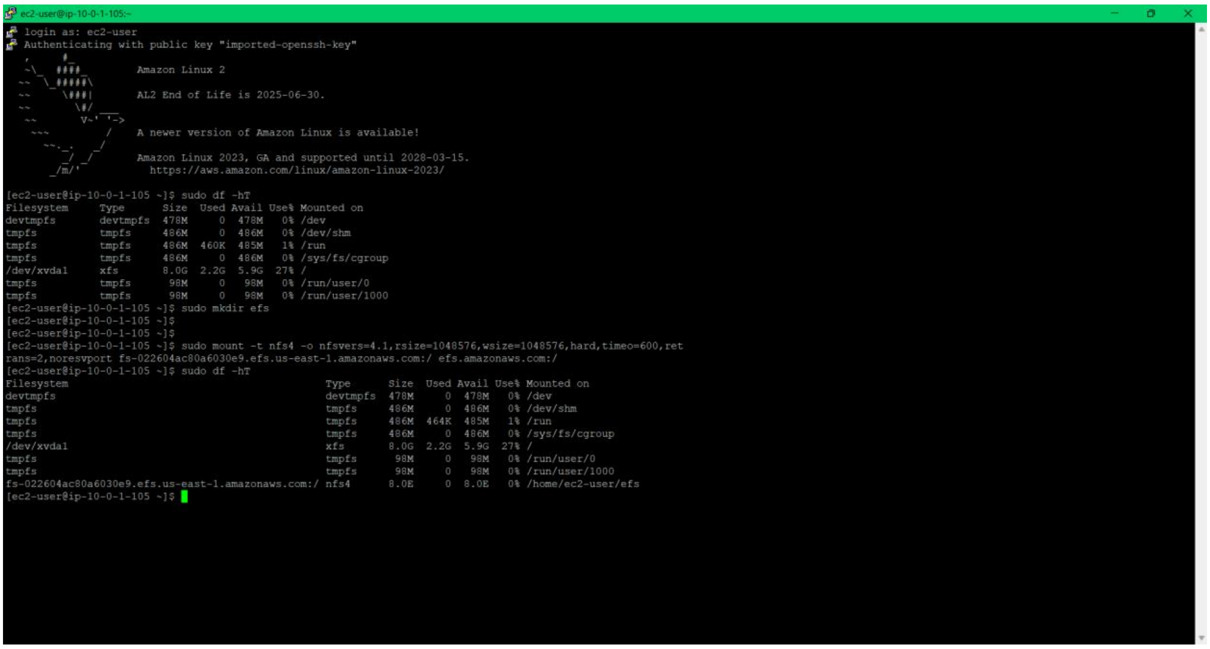




In my SSH session, I create a new directory named "efs" using the command sudo mkdir efs. I go back to the AWS Management Console and access the EFS service, there I select the specific EFS file system, such as "My First EFS File System". In the EFS Console, I choose "Attach" to open the instructions for mounting the EFS file system to an EC2 instance. I copy the entire mount command provided in the "Using the NFS client" section.

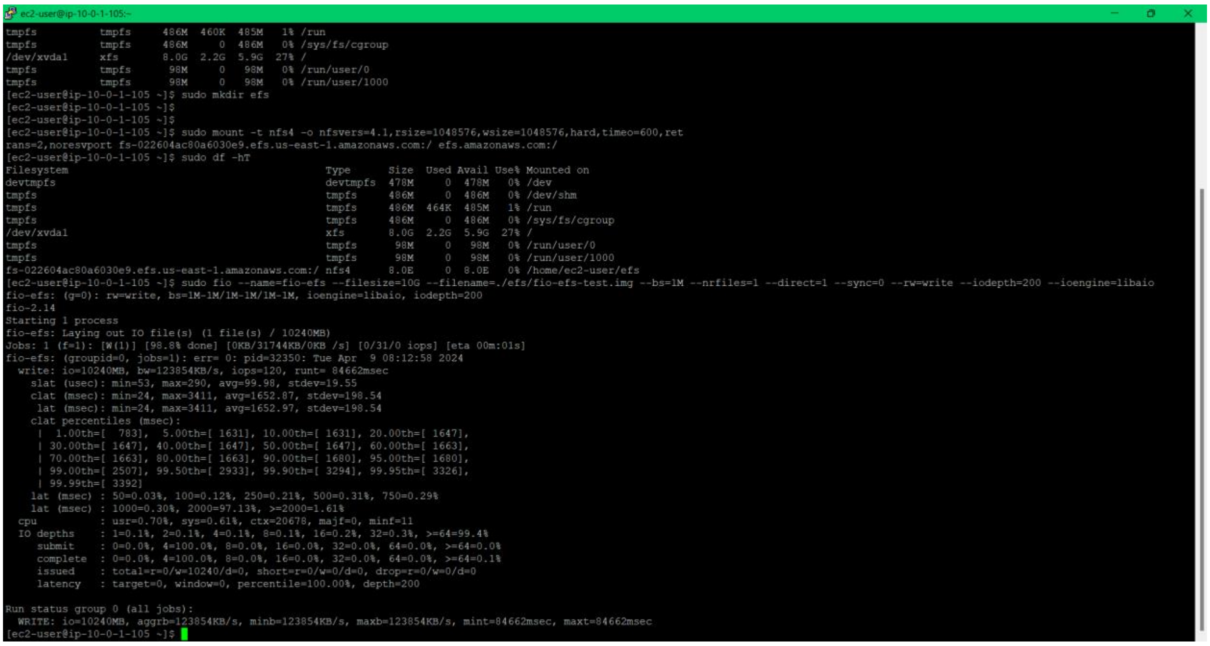
Back in my SSH session on the Linux EC2 instance, I paste the copied mount command and press ENTER to execute it. This mounts the EFS file system to the EC2 instance.

To get a summary of available and used disk space, I enter sudo df -hT in the SSH session. This command displays information about the mounted EFS file system, including its type and size.



Task 5: Examining the performance behavior of your new EFS file system

I just booted up my Amazon EC2 instance. A tool named fio is already installed. I'm going to run a test. The test should take around 5-10 minutes to complete. Once it's done, I'll take a look at the output to see the exact write speed.



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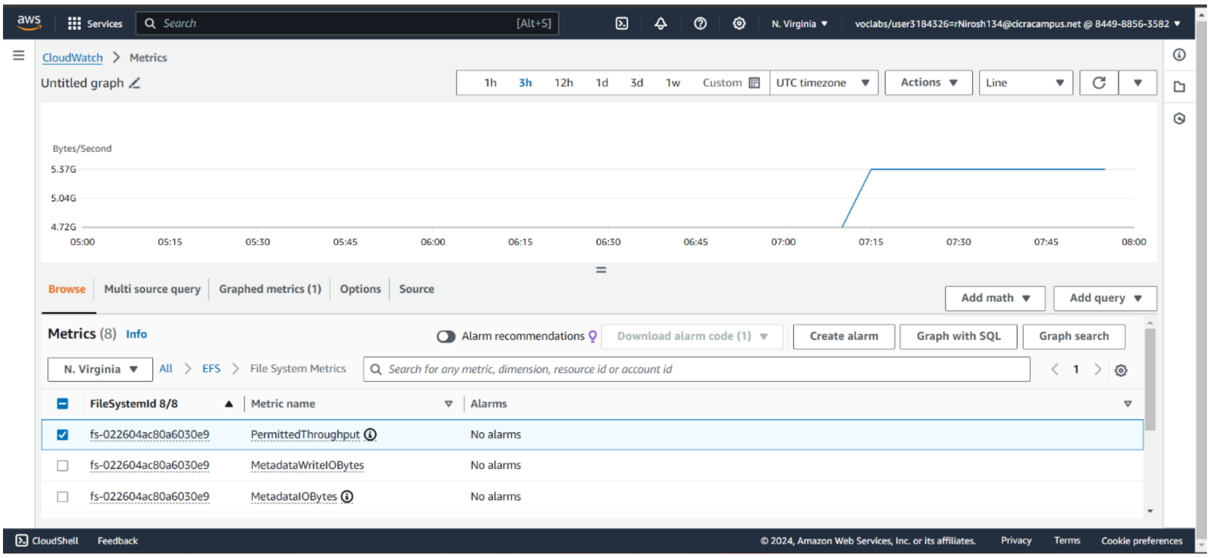
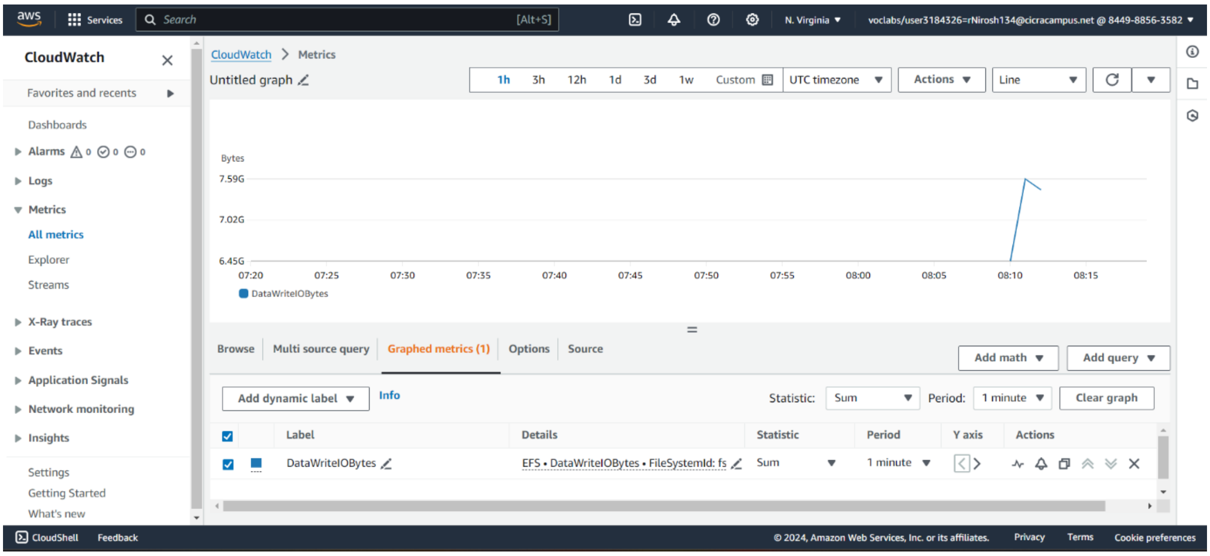
Monitoring performance by using Amazon CloudWatch



I navigated to the EFS metrics section and searched for a metric called "PermittedThroughput." It might take a few minutes for the data to populate.

After looking at the permitted throughput, I switched to another metric called "DataWriteIOBytes." This shows the actual write activity during a specific time period.



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Grade:

My Grade is 15/15